PEDAGOGICAL SCIENCES

HYPOTHESIS TESTING: PEDAGOGICAL EXPLANATION OF TYPE I AND TYPE II ERRORS

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Abstract

Hypothesis testing is a core tool of statistical inference, allowing researchers to evaluate population parameters using data. This paper outlines key concepts such as null and alternative hypotheses, test statistics, rejection regions, significance levels, and p-values. It reviews common tests, including z-tests and t-tests, and illustrates applications in quality control, claims testing, and comparing proportions. Emphasis is placed on sampling variability, decision-making, and the proper use of statistical methods in research.

Keywords: Hypothesis testing, Null hypothesis, Alternative hypothesis, Type I error (α error), Type II error (β error), Significance level (α), p-value

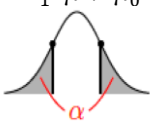
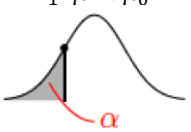
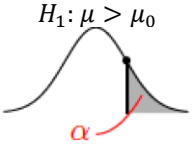
Introduction

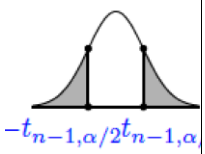
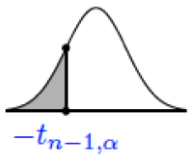
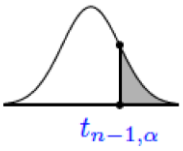
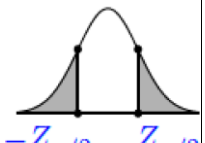
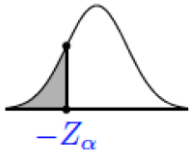
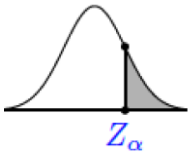
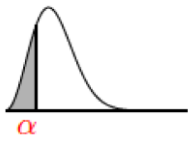
Hypothesis testing is one of the core methodologies of inferential statistics, providing a structured framework for decision-making under uncertainty. The procedure relies on formulating two competing statements: the null hypothesis, representing the default assumption, and the alternative hypothesis, representing the rival claim. Evidence from sample data is then evaluated to determine whether the null hypothesis should be rejected. Central to this framework are the notions of **Type I error**-rejecting a true null hypothesis-and **Type II error**-failing to reject a false null hypothesis. The probabilities of these errors, denoted by α and β respectively, together with the concept of the **power of a test** (1 – β), play a crucial role in designing efficient and reliable statistical tests.

Literature Review

This [1] classic book by R. A. Fisher (1925) laid the foundation for modern statistical inference, introducing the concept of *significance testing*. The seminal paper by Neyman and Pearson [2], [3], [4] formalized the theory of hypothesis testing. They introduced the concepts of **null and alternative hypotheses**, **Type I and Type II errors**, and developed the **Neyman–Pearson lemma**, which establishes the most powerful test for a given size α. This work remains a cornerstone of statistical testing theory. Casella and Berger's *Statistical Inference* (2002) [5] provides a rigorous mathematical treatment of hypothesis testing, including likelihood ratio tests, large-sample approximations, and power analysis. Lehmann and Romano's *Testing Statistical Hypotheses* (2005) [6] remains one of the most comprehensive treatments of the subject, extending classical results and integrating modern advances in statistical testing. Efron and Tibshirani (1994) [7] introduced the bootstrap as a resampling-based alternative to traditional parametric testing, while Davison and Hinkley (1997) [8] provided a systematic review of bootstrap and simulation-based methods, linking them to hypothesis testing and error probabilities.

Educational resources also play a role in disseminating these ideas. Aliev's text (2009) [9] and [10] illustrate the practical application of hypothesis testing through solved problems, with emphasis on Type I and II errors, test construction, and interpretation.

№		A two tailed test	A left tailed test	A right tailed test	T.S.	Table Aliyev
1		$H_0: \theta = \theta_0$ $H_1: \theta \neq \theta_0$	$H_0: \theta \geq \theta_0 (\theta = \theta_0)$ $H_1: \theta < \theta_0$	$H_0: \theta \leq \theta_0 (\theta = \theta_0)$ $H_1: \theta > \theta_0$		
2	Test the mean(μ) of population that is normally distributed population variance known	$H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$ 	$H_0: \mu = \mu_0$ $(\mu \geq \mu_0)$ $H_1: \mu < \mu_0$ 	$H_0: \mu = \mu_0$ $(\mu \leq \mu_0)$ $H_1: \mu > \mu_0$ 	$T.S. = z = \frac{\underline{x} - \mu}{\sigma/\sqrt{n}}$	2

3	(μ) variance unknown. Large sample size	-//-	-//-	-//-	$z = \frac{\underline{x} - \mu}{s/\sqrt{n}}$	
4	(μ) variance unknown. Small sample size n <30	-//- 	-//- 	-//- 	$t_{n-1} = \frac{\underline{x} - \mu}{s/\sqrt{n}}$	4
5	(p) Large sample size	$H_0: p = p_0$ $H_1: p \neq p_0$ 	$H_0: p = p_0$ $(p \geq p_0)$ $H_1: p < p_0$ 	$H_0: p = p_0$ $(p \leq p_0)$ $H_1: p > p_0$ 	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	2
6	σ^2	$H_0: \sigma^2 = \sigma_0^2$ $H_1: \sigma^2 \neq \sigma_0^2$ 	$H_0: \sigma^2 = \sigma_0^2$ $(\sigma^2 \geq \sigma_0^2)$ $H_1: \sigma^2 < \sigma_0^2$ 	$H_0: \sigma^2 = \sigma_0^2$ $(\sigma^2 \leq \sigma_0^2)$ $H_1: \sigma^2 > \sigma_0^2$ 	$\chi^2_{n-1} = \frac{(n-1)\cdot s^2}{\sigma^2}$	3
7	D_0 Tests based on paired samples	$H_0: \mu_x - \mu_y = D_0$ $H_1: \mu_x - \mu_y \neq D_0$	$H_0: \mu_x - \mu_y = D_0$ $(\mu_x - \mu_y \geq D_0)$ $H_1: \mu_x - \mu_y < D_0$	$H_0: \mu_x - \mu_y = D_0$ $(\mu_x - \mu_y \leq D_0)$ $H_1: \mu_x - \mu_y > D_0$	$T.S. = \frac{\underline{d} - D_0}{s_{\underline{d}}/\sqrt{n}}$	4
8	D_0 Based on independent samples. (Known variance)	-//-	-//-	-//-	$z = \frac{(\underline{x} - \underline{y}) - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}}$	2
9	D_0 Based on independent samples	-//-	-//-	-//-	$t = \frac{(\underline{x} - \underline{y}) - (\mu_x - \mu_y)}{\sqrt{\frac{s_p^2}{n_x} + \frac{s_p^2}{n_y}}}$	4

	(Population variances are unknown and equal)				$s_p^2 = \frac{(n_x-1)\cdot s_x^2 + (n_y-1)\cdot s_y^2}{(n_x+n_y-2)}$	
10	The difference between two population proportions	$H_0: p_x - p_y = 0$ $H_1: p_x - p_y \neq 0$	$H_0: p_x - p_y = 0$ $(p_x - p_y \geq 0)$ $H_1: p_x - p_y < 0$	$H_0: p_x - p_y = 0$ $(p_x - p_y \leq 0)$ $H_1: p_x - p_y > 0$	$Z = \frac{\left(\hat{p}_x - \hat{p}_y\right) - \left(p_x - p_y\right)}{\sqrt{\frac{p_x(1-p_y)}{n_x} + \frac{p_y(1-p_y)}{n_y}}}$	2
10	Large sample size	-//-	-//-	-//-	$Z = \frac{\left(\widehat{p_x} - \widehat{p_y}\right) - (p_x - p_y}{\sqrt{\frac{p_0(1-p_0)}{n_x} + \frac{p_0(1-}{n_y}}}$ Where $p_0 = \frac{n_x \cdot \widehat{p_x} + n_y \cdot \widehat{p_y}}{n_x + n_y}$ Or $p_0 = \frac{x_1 + y_1}{n_x + n_y}$	2

A statistical test of hypothesis procedure contains the following five steps:

1. State the null and alternative hypothesis
2. Select the distribution to use
3. Determine the rejection and nonrejection regions
4. Calculate the value of the test statistic
5. Make a decision

Concepts of hypothesis testing

Let us consider an example about coffee cans:

A company may claim that, on average, its cans contain 100 grams of coffee. A government agency may want to test whether or not such cans contain, on average, 100 grams of coffee. Suppose we take a sample of 50 cans of the coffee under investigation. We then find out that the mean amount of coffee in these 50 cans is 97 grams. Based on these results, can we state that on average, all such cans contain less than 100 grams of coffee and that the company is lying to the public? Not until we perform a test of hypothesis. The reason is that the mean $\underline{x} = 97$grams are obtained from the sample. The difference between 100 grams (the required amount for the population) and 97 grams (the observed average amount for the sample) may have occurred only because of the sampling error. Another sample of 100 cans may give us a mean of 105 grams. Therefore, we make a test of hypothesis to find out how large the difference between 100 grams and 97 grams is and to investigate whether or not this difference has occurred as a result of chance alone. If 97 grams is the mean of all cans and not for only 100 cans, then we do not need to make a test of hypothesis. Instead, we can immediately state that the mean amount of coffee in all such cans is less than 100 grams. We perform a test of hypothesis only when we are making a decision about a population parameter based on the value of a sample statistic.

Example:

1) A manufacturer of detergent claims that the content of boxes sold weigh on average at least 160 grams. The distribution of weights is known to be normal, with standard deviation of 14 grams. A random sample of 16 boxes yielded a sample mean weight of 158.9 grams. Test at the 10% significance level the null hypothesis that the population mean is at least 160 grams.

Solution:

Let $\underline{x}$ be the mean average of all boxes and n =16; σ =14; $\underline{x}$ =158.9; α = 0.1

$$H_0: \mu \geq 160$$
$$H_1: \mu < 160$$

P(Z>$z_{0.1}$) = 0.1 ; F($z_{0.1}$) = 0.9; $z_{0.1} = -1.285$;

$$T.S. = z = \frac{\underline{x} - \mu}{\sigma/\sqrt{n}} = \frac{158.9 - 160}{14/\sqrt{16}} = -0.31$$

In the final step we make a decision based on the value of the test statistic falling in the nonrejection region. Hence we accept H_0 and conclude that based on sample information, it appears that the mean weight of all boxes is greater than 160 grams.

2) Variance of yearly earnings of all state employees for all 40 states is $490000 square dollars. A sample of 29 employees selected from state A produced a variance of their earnings equal to $600 000 square dollars. Test

at 5% significance level if the variance of yearly earnings of state employees in state A is different from $490 000 square dollars. Assume that the yearly earnings of all state employees in state A have an (approximate) normal distribution.

Solution:

From the given information, n = 29; α =0.05 ; s^2 =600000; $\sigma^2 = 490000$

The null and alternative hypotheses are

$$H_0: \sigma^2 = 490000$$
$$H_1: \sigma^2 \neq 490000$$

$\alpha/2$ =0.025; We use Chi square distribution $\chi^2_{n-1,\frac{\alpha}{2}} = \chi^2_{28,0.025} = 44.461$; $\chi^2_{n-1,1-\frac{\alpha}{2}} = \chi^2_{28,0.975} = 15.308$

$$TS = \chi^2_{n-1,1-\frac{\alpha}{2}} = \frac{(n-1)s^2}{\sigma^2} = \frac{(29-1)\cdot 600000}{490000} = 34.286$$

The value of the test statistic 34.286 is between the two critical values and falls in the nonrejection region. Consequently we fail to reject H_0 and conclude that the population variance of yearly earnings of all employees in state A is not different from 490000 square dollars.

Type I error

A type I error occurs when a true null hypothesis is rejected. The value α represents the probability of committing this type of error, that is α=$P(H_0$ is rejected / H_0 is true)

The value αrepresents the significance level of the test.

Type II error

A Type II error occurs when a false null hypothesis is not rejected. The value β represents the probability of committing a Type II error, that is β=$P(H_0$ is not rejected / H_0 is false) The value (1−β) is called the power of the test. It represents the probability of not making a Type II error.



Remark: Note that the null hypothesis always has an equal to (=) or a less or equal to (≤) or a greater than or equal to (≥) sign and the alternative hypothesis always has a not equal to (≠) or a greater than (>) or a less than (<) sign.

Hypothesis testing using the *p* –value approaches

Definition:

The p-value is the smallest significance level at which the null hypothesis is rejected. Using the p-value approach, we reject the null hypothesis if p- value<α and we do not reject the null hypothesis if p- value ≥α

Example: The management of Health club claims that its members lose an average of 10kg or more within the first month after joining the club. A random sample of 36 members of this health club was taken and found that they lost an average of 9.2 kg within the first month of membership with standard deviation of 2.4kg. Find the *p*- value for this test.

Solution:

Let μ be the mean weight lost during the first month of membership by all members and $\underline{x}$ be the corresponding mean for the sample.



Step 1. *State the null and alternative hypothesis*

$H_0: \mu \geq 10$ (The mean weight lost is 10kg or more)

$H_1: \mu < 10$ (The mean weight lost is less than 10kg)

Step 2. *Select the distribution to use*

Because the sample size is large we use the normal distribution to make the test and calculate *p*-value.

Step 3. Calculate the *p*-value.

The < sign in the alternative hypothesis indicates that the test is left tailed. The *p*- value is given by the area in the left tail of the sampling distribution curve.

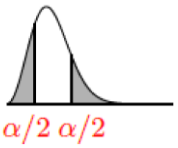
To find this area, we first find the *z* value

$$T.S. = z = \frac{\underline{x} - \mu}{s/\sqrt{n}} = \frac{9.2 - 10}{2.4/\sqrt{36}} = -2.00$$

Area to the left F(-2.00)=1-F(2.00)=1-0.9772=0.0228

Thus, based on the *p*- value of 0.0228 we can state that for any α (significance level) greater than 0.0228 we will reject the null hypothesis and for any α less than 0.0228 we will accept the null hypothesis.

Suppose we make the test for this example at α = 0.01. Because α =0.01is less than *p*-value of 0.0228, we will not reject the null hypothesis. Now suppose we make the test a t α = 0.05. Because α = 0.05 is greater than the *p*-value of 0.0228, we will reject the null hypothesis.



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